

# Adapting Vision-Language-Action Models for BEHAVIOR-1K Household Tasks

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**Abstract.** BEHAVIOR-1K is a long-horizon embodied AI benchmark in which an agent must solve realistic household activities involving navigation, object interaction, and manipulation.

$\pi_{0.5}$ -based VLA solution for BEHAVIOR-1K. Avg  $Q$  0.365, +40% over leaderboard #1. We evaluated and adapted vision-language-action (VLA) policies on six selected BEHAVIOR-1K tasks: picking up trash, picking up toys, tidying bedroom, collecting children’s toys, sorting household items, and cleaning up a desk. Our main branch follows a Robot Learning Collective-style  $\pi_{0.5}$  policy, using task embeddings, stage tracking, correlated-noise flow matching, and rejection-sampling fine-tuning (RFT) with stage-guided perturbations. Our best six-task average  $Q$ -score is 0.344, with the strongest single-task results on collecting children’s toys (0.604) and picking up trash (0.567). We also tested two diagnostic branches: a second-checkpoint fine-tuning setting with point-cloud input, and NVIDIA GR00T-N1.6-BEHAVIOR1k. The ablations show that 3D information can improve the second-checkpoint branch, while GR00T’s 200-demonstration fine-tuning often degrades task progress despite preserving partial navigation behavior. Overall, our results suggest that BEHAVIOR-1K performance is limited less by semantic recognition and more by data coverage, stage robustness, and reliable grasp-and-place execution.

**Keywords:** BEHAVIOR-1K · Vision-Language-Action model · robot learning · flow matching · rejection-sampling fine-tuning

## 1 Introduction

BEHAVIOR-1K is a human-centered embodied AI benchmark containing 1,000 everyday activities in realistic simulated environments [1]. The 2025 BEHAVIOR Challenge selects 50 full-length household tasks and evaluates policies in interactive scenes with diverse objects, physical interactions, and long-horizon goal conditions [2]. In contrast to short tabletop manipulation, these tasks require a policy to chain navigation, perception, bimanual manipulation, and task-stage reasoning over many time steps. A typical episode may require the robot to move to the right room, localize multiple objects, grasp them without dropping them, transport them, and place them in a target receptacle or region.

The challenge metric used throughout this report is  $Q$ -score, which gives partial credit according to the fraction of goal predicates satisfied by the policy. This metric is more informative than binary success when agents make meaningful progress but fail to complete all subgoals. Our project focuses on six tasks listed in Table 1. Task 1, `picking_up_trash`, is treated as the main task, while the other tasks test whether the same VLA adaptation ideas transfer to toy collection, tidying, sorting, and desk cleanup.

Our goal is not to train a robot policy from scratch. Instead, we ask how far a strong pretrained VLA foundation model can be pushed under a course-project setting. We therefore build on recent BEHAVIOR Challenge solutions and test three implementation branches: (1) an RFT branch based on the Robot Learning Collective model, (2) a second-checkpoint branch with 10k-step supervised fine-tuning and point-cloud input, and (3) a GR00T branch used to analyze zero-shot versus low-data fine-tuning behavior.

## 2 Problem Setting

The selected tasks share three properties that make them difficult for current VLA policies. First, they are long-horizon: the robot must execute hundreds or thousands of low-level actions before the final predicates are satisfied. Second, they are partially ambiguous: the same observation may correspond to different task

**Table 1.** Selected BEHAVIOR-1K tasks used in our project.

| Theme               | Task ID | Task name                              |
|---------------------|---------|----------------------------------------|
| Trash pickup        | 1       | <code>picking_up_trash</code>          |
| Toy collection      | 7       | <code>picking_up_toys</code>           |
| Tidying and sorting | 18      | <code>tidying_bedroom</code>           |
| Toy collection      | 21      | <code>collecting_childrens_toys</code> |
| Tidying and sorting | 27      | <code>sorting_household_items</code>   |
| Tidying and sorting | 29      | <code>clean_up_your_desk</code>        |

stages, such as approaching an object, carrying it, or placing it into a target area. Third, success depends on physical execution rather than language understanding alone. If the robot reaches the correct object but fails to close the gripper, the formal  $Q$ -score may remain low.

We distinguish *task progress* from  $Q$ -score. Task progress is a denser diagnostic measure that indicates whether the robot is moving through meaningful sub-stages.  $Q$ -score is stricter because it checks the actual BEHAVIOR goal predicates. This distinction is important in our GR00T experiments: the model can navigate and approach objects, giving nonzero progress, while still obtaining 0.0  $Q$ -score when grasp-and-place execution fails.

### 3 Related Work

We organize the related work into three points, one for each reference our project builds on: the first-place Robot Learning Collective solution, the second-place Openpi Comet solution, and the NVIDIA GR00T BEHAVIOR example we use as a diagnostic baseline.

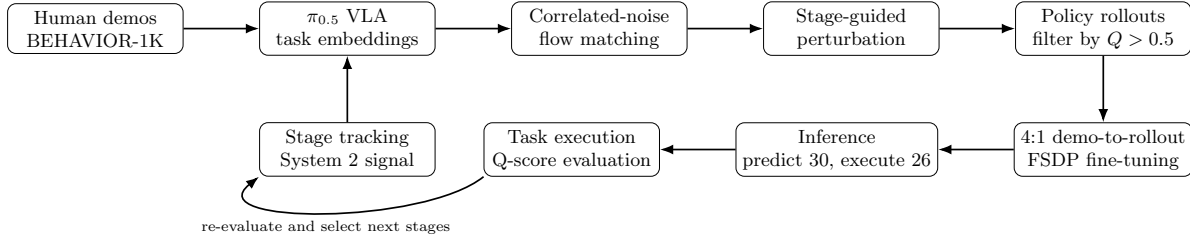
- (1) **First-place Robot Learning Collective.** It adapts  $\pi_{0.5}$  with correlated noise for flow matching, System 2 stage tracking, multi-sample training, action inpainting, and challenge-specific inference corrections [3]. Its key observation is that action sequences are correlated across time and joints, so the noise model and inference procedure should preserve this correlation rather than treat each action dimension independently.
- (2) **Second-place Openpi Comet.** Comet [4] also builds on  $\pi_{0.5}$  but focuses on scaling data and post-training. Broadening pre-training from 7 to 10 and then all 50 challenge tasks progressively unlocks rare, compositional activities (validation Q 0.192), and motion-planner plus offline-RL trajectories augment the human demonstrations. Instead of sample-inefficient online RL it adopts rejection-sampling fine-tuning (RFT): perturb the initial pose, roll out the policy, and keep successful episodes as an iterative offline data flywheel, reaching validation Q 0.345 with an oracle ceiling of 0.611.
- (3) **NVIDIA GR00T-N1.6-BEHAVIOR1k example.** The official Isaac-GR00T example ships the `nvidia/GR00T-N1.6-BEHAVIOR1k` checkpoint with scripts for fine-tuning under the BEHAVIOR R1 PRO embodiment and a server-client rollout pipeline [6]. Unlike the  $\pi_{0.5}$  solutions it is not meant to be best-performing; we use it only to test whether low-data fine-tuning helps or harms a pretrained foundation model.

Overall, we adopt the first-place design for stage awareness and correlated-noise flow matching together with the second-place RFT recipe for post-training. We further believe the Comet line still has clear headroom: the gap between its 0.345 validation score and the 0.611 oracle ceiling suggests that continued fine-tuning with more RFT rounds and data can keep improving performance.

## 4 Method

### 4.1 Main RFT Branch

The main branch uses a Robot Learning Collective-style VLA policy. The backbone is based on  $\pi_{0.5}$ , which combines a PaliGemma-style VLM component with an action expert. The action space is 23-dimensional



**Fig. 1.** Overview of our main branch. Stage tracking and correlated-noise flow matching come from the first-place style policy, while stage-guided perturbation and accepted-rollout mixing follow the RFT idea used by Comet.

delta joint position control. Instead of relying only on language prompts, the model uses 50 trainable task embeddings, one per challenge task. The action head is a flow-matching model with correlated noise, parameterized in our setting with  $\beta = 0.5$ . At inference time, the policy predicts 30 future time steps and executes 26, leaving a four-step overlap as an anchor for the next prediction window.

Figure 1 summarizes our full pipeline. The pretraining side provides the task-conditioned VLA policy and its stage-aware flow-matching action head. The post-training side perturbs states around different task stages, runs the policy, keeps rollouts whose  $Q$ -score exceeds 0.5, and mixes these accepted rollouts with human demonstrations. The final policy is evaluated again on the selected BEHAVIOR-1K tasks.

Our RFT procedure has four steps. First, perturbations are injected at different task stages rather than only at the global task start. Second, the current policy collects rollouts and only trajectories with  $Q > 0.5$  are kept. Third, the accepted trajectories are converted to Parquet format with per-timestamp normalization. Finally, the model is fine-tuned with fully sharded data parallelism on 2–3 GPUs using a 4:1 ratio of human demonstrations to policy rollouts. The purpose of this design is to allocate more training signal to stage transitions where long-horizon policies frequently fail.

## 4.2 Minor Branches

The first minor branch uses a second checkpoint and applies 10,000-step supervised fine-tuning on the six selected task instances. We vary two factors: whether point-cloud input is provided and whether the inference noise distribution matches the training setting. This branch tests whether adding explicit 3D information is useful for BEHAVIOR manipulation.

The second minor branch evaluates NVIDIA GR00T-N1.6-BEHAVIOR1k. We compare zero-shot task progress against several 200-demonstration fine-tuning variants. This branch is mainly diagnostic: it reveals whether low-data fine-tuning improves a pretrained VLA prior or instead overfits the action head. In our implementation, the GR00T VLM backbone is frozen by default, so the measured degradation primarily reflects the action head, projector, and their coupling, rather than full-model language-vision forgetting.

## 5 Experiments

### 5.1 Evaluation Protocol

We evaluate each policy by mean  $Q$ -score over repeated trials. Most tasks use 8–10 trials; Task 29 has only two trials due to time and compute limitations. This should be interpreted carefully: our six-task average is an internal subset evaluation, not a full official 50-task leaderboard submission. Nevertheless, the subset is useful because it includes trash pickup, toy collection, bedroom tidying, object sorting, and desk cleanup, which exercise navigation, grasping, placement, and multi-object interaction.

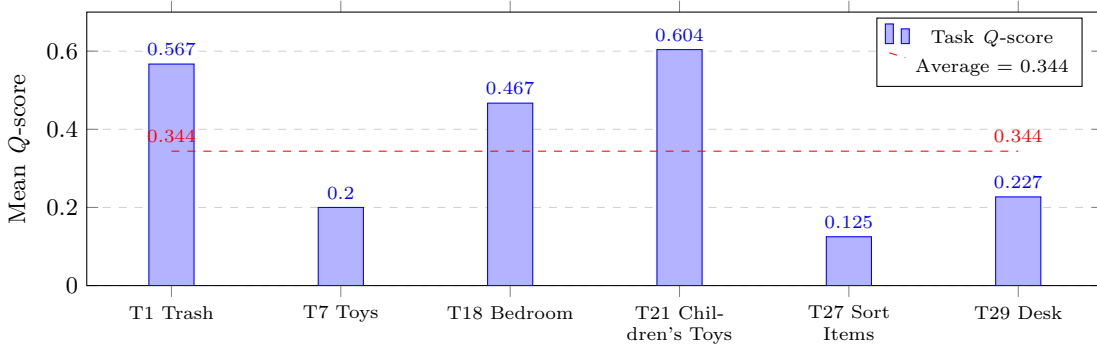
### 5.2 Main Results

Table 2 shows the main branch results. The average  $Q$ -score over the six tasks is 0.344. The strongest task is collecting children’s toys with mean  $Q$ -score 0.604 over eight trials. The main task, picking up trash, reaches 0.567 over ten trials. Tidying bedroom also performs well at 0.467. The hardest task in our subset is

**Table 2.** Mean  $Q$ -score of our main branch on six BEHAVIOR-1K tasks.

| Task ID | Task name                 | Trials | Mean $Q$ -score |
|---------|---------------------------|--------|-----------------|
| 1       | picking_up_trash          | 10     | 0.567           |
| 7       | picking_up_toys           | 10     | 0.200           |
| 18      | tidying_bedroom           | 10     | 0.467           |
| 21      | collecting_childrens_toys | 8      | 0.604           |
| 27      | sorting_household_items   | 10     | 0.125           |
| 29      | clean_up_your_desk        | 2      | 0.227           |
| Average |                           |        | 0.344           |

sorting household items with 0.125, suggesting that object-category reasoning and correct placement remain unstable.



**Fig. 2.** Task-wise  $Q$ -score of the main branch. The dashed line denotes the six-task average, 0.344. This figure is generated directly in LaTeX so it appears without requiring an external image file.

Figure 2 illustrates the spread across tasks. The high scores on Task 1 and Task 21 show that the policy can make meaningful progress in pickup and collection tasks. However, the weaker scores on Task 7, Task 27, and Task 29 indicate that long-horizon performance is still brittle when multiple object categories and final placement constraints are involved. In particular, sorting household items requires both recognizing object categories and placing them into the correct destination. A policy can navigate to an object but still lose most of the score if the final relation predicate is not satisfied.

## 6 Ablation and Diagnostic Results

### 6.1 Second Checkpoint with Point-Cloud Input

Table 3 reports the second-checkpoint branch. Without point-cloud input, both the baseline checkpoint and the 10k-step supervised fine-tuned checkpoint obtain 0.000 average  $Q$ -score. Adding point-cloud input with mismatched task-correlated noise also fails. The only nonzero setting is 10k-step supervised fine-tuning with Gaussian noise and point-cloud input, reaching 0.100 average  $Q$ -score over ten trials. This supports two practical conclusions: 3D information helps, and the inference noise distribution must match the training distribution.

The result is small but informative. Point clouds do not solve the task alone, yet they provide geometric structure that RGB-only observations may miss, especially around occlusions, object-receptacle distance, and grasp approach direction. The failed task-correlated-noise setting also shows that pretraining and inference must be distributionally aligned. If the model is trained under one noise structure and sampled under another, the denoised action chunk can become inconsistent, which is damaging in long-horizon tasks where small action errors accumulate.

**Table 3.** Ablation of the second-checkpoint branch. PCD denotes point-cloud input.

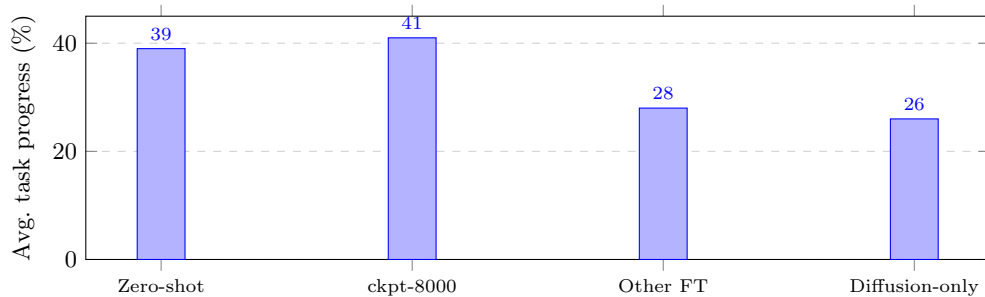
| Setting       | Noise     | PCD input | Trials | Avg. $Q$ -score |
|---------------|-----------|-----------|--------|-----------------|
| Baseline pt50 | Gaussian  | No        | 3      | 0.000           |
| SFT 10k       | Gaussian  | No        | 3      | 0.000           |
| SFT 10k       | task-corr | Yes       | 5      | 0.000           |
| SFT 10k       | Gaussian  | Yes       | 10     | <b>0.100</b>    |

**Table 4.** GR00T diagnostic results on Task 1. Progress and  $Q$ -score measure different levels of completion.

| Configuration                   | Avg. progress            | $Q$ -score |
|---------------------------------|--------------------------|------------|
| Zero-shot                       | 39% ( $n = 20$ )         | 0.0        |
| Full fine-tune, ckpt-8000       | $\sim 41\%$ ( $n = 20$ ) | 0.0        |
| Full fine-tune, ckpt-6000/10000 | 27–29%                   | 0.0        |
| Diffusion-only                  | 26%                      | 0.0        |
| Curriculum, first third demos   | 27%                      | 0.0        |
| Approach-only, 80 frames        | 26%                      | 0.0        |
| Approach-only, gripper cutoff   | 29%                      | 0.0        |

## 6.2 GR00T Fine-Tuning Analysis

The GR00T branch reveals a different failure mode. The pretrained zero-shot GR00T policy reaches about 39% average task progress on Task 1. The best fine-tuned checkpoint, ckpt-8000, reaches about 41%, which is only a tie-level improvement. Other fine-tuned variants, including ckpt-6000, ckpt-10000, diffusion-only training, curriculum-style truncation, and approach-only variants, drop to roughly 26–29% progress. Importantly, all of these GR00T settings have  $Q$ -score 0.0 in our trials, meaning that partial progress does not translate into completed goal predicates.

**Fig. 3.** GR00T fine-tuning diagnostic on Task 1. The best 200-demo fine-tuned checkpoint only approximately matches zero-shot progress, while most fine-tuned variants degrade. All reported GR00T settings still have 0.0  $Q$ -score in our runs.

These results suggest that low-data fine-tuning can hurt more than help. GR00T’s VLM backbone is frozen by default, so the degradation is unlikely to be catastrophic forgetting in semantic perception. Instead, the degradation is concentrated in the action head and its coupling with the projector. With only 200 demonstrations, the fine-tuned policy does not receive enough coverage to improve long-horizon grasping and placement, but it can still distort the pretrained manipulation prior.

## 7 Discussion

Our experiments point to three main insights.

*Fine-tuning with too few demonstrations is risky.* The GR00T experiment shows that a pretrained VLA policy may already contain broad manipulation priors. Fine-tuning on 200 demonstrations does not necessarily add enough task-specific coverage to improve the policy, but it can still overfit the action head. This is consistent with the larger-scale BEHAVIOR solutions, which rely on much more data, accepted rollouts, planner trajectories, or explicit recovery strategies rather than a small single-task imitation set.

*Task progress and  $Q$ -score should be interpreted separately.* Policies can navigate toward relevant objects and still obtain zero  $Q$ -score if grasping, carrying, or placement fails at the final predicates. This is why GR00T can reach nearly 40% progress while still obtaining 0.0  $Q$ -score: the robot begins the task correctly but does not satisfy the formal BDDL goal conditions. For long-horizon household tasks, the hard part is often not the first approach motion but the reliable completion of manipulation relations.

*Stage robustness, geometry, and distribution matching are central.* The main RFT branch improves performance by perturbing and collecting rollouts at different stages, which increases data coverage around failure-prone transitions. The second-checkpoint branch shows that point-cloud input can help, but only when the inference-time noise process matches the training distribution. These findings suggest that future BEHAVIOR policies should not treat perception, action sampling, and rollout collection as independent modules. They must be jointly aligned so that the model sees the same state and noise distributions during training and deployment.

A limitation of our report is that the six-task average is not directly comparable to the full official leaderboard, which evaluates 50 tasks and uses a fixed public/held-out split. Our score should therefore be interpreted as a strong internal result on a selected subset rather than an official ranking. Another limitation is the small number of trials for Task 29. More trials and a wider task set would be needed for a statistically stable conclusion.

## 8 Conclusion

We implemented and analyzed three VLA-based approaches for BEHAVIOR-1K. The main RFT branch, built on a  $\pi_{0.5}$ -style policy with task embeddings, stage tracking, correlated-noise flow matching, and stage-guided perturbation, obtains an average  $Q$ -score of 0.344 on six selected tasks. The best individual results are 0.604 on collecting children’s toys and 0.567 on picking up trash. The second-checkpoint branch shows that point-cloud input can help, while the GR00T branch shows that 200-demonstration fine-tuning is often insufficient and can degrade pretrained action priors.

The main takeaway is that current VLA models already contain useful navigation and manipulation priors, but BEHAVIOR-1K requires more than partial progress. Reliable success depends on dense stage coverage, recovery from off-distribution states, correct action-noise modeling, and robust grasp-and-place execution.

## 9 Future Direction

Future work should expand RFT rollout coverage, perturb states at automatically detected failure stages, use richer subgoal supervision, and test parameter-efficient adaptation that preserves the pretrained policy. We also note a hardware caveat: on Blackwell GPUs (compute capability 12.0) the simulator’s RTX renderer produced noisy renders that silently corrupted the visual input.

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